

LANCASTER FOX, CA, USA

WMO: 723816

Lat: 34.741N

Lon: 118.212W

Elev: 2338

StdP: 13.50

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 03159

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	21.3	24.9	-2.7	5.2	39.1	1.4	6.4	42.8	32.9	50.9	28.9	51.6	3.0	270	0.631

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	29.0	103.5	65.7	100.9	64.5	98.6	63.6	68.6	95.9	67.2	94.9	65.8	93.7	13.9	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
59.0	81.2	80.1	55.9	72.6	80.9	53.6	66.5	79.6	34.0	94.9	32.8	95.0	31.6	93.4	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
30.2	27.1	25.0	DB	15.2	108.9	4.0	1.9	12.3	110.3	10.0	111.4	7.7	112.4	4.8	113.8	
			WB	12.1	71.1	3.7	1.9	9.4	72.4	7.3	73.5	5.2	74.6	2.6	75.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	62.5	45.5	48.1	54.0	59.1	67.6	76.9	83.2	81.8	74.4	62.7	51.6	44.1
	DBStd	15.03	5.73	5.89	6.45	7.28	8.05	7.56	4.91	4.54	6.07	6.58	6.71	6.01
	HDD50	564	160	97	34	10	0	0	0	0	0	2	60	201
	HDD65	2883	604	475	343	204	69	9	0	0	7	122	402	648
	CDD50	5125	21	43	158	283	544	806	1030	987	731	396	109	17
	CDD65	1968	0	0	3	26	148	365	565	522	287	51	1	0
	CDH74	28071	1	14	177	567	1967	4829	7830	7312	4115	1155	104	0
	CDH80	15814	0	1	30	176	909	2744	4766	4462	2281	436	9	0
Wind	WSAvg	10.9	7.4	9.5	11.5	14.1	15.0	14.8	12.9	11.4	9.8	9.1	8.1	7.7
Precipitation	PrecAvg	7.5	1.2	1.9	1.4	0.4	0.1	0.0	0.1	0.1	0.2	0.3	0.5	1.1
	PrecMax	14.4	4.6	6.2	5.6	1.8	1.1	0.8	0.9	1.0	1.3	2.4	2.1	5.4
	PrecMin	1.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	3.6	1.1	2.1	1.5	0.5	0.3	0.2	0.2	0.2	0.4	0.5	0.6	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	70.3	75.8	84.3	90.2	98.1	105.5	107.6	105.9	101.1	93.0	81.3	70.1
		MCWB	49.8	51.1	55.0	58.5	62.5	65.0	67.7	67.5	64.6	60.2	53.9	51.4
	2%	DB	66.3	71.9	79.0	85.0	93.4	100.4	104.1	102.8	98.4	88.9	77.2	65.3
		MCWB	47.4	50.0	53.6	55.9	60.4	63.3	66.2	65.6	63.5	58.2	52.1	48.1
	5%	DB	63.2	67.3	74.9	80.9	89.6	97.3	101.1	100.3	95.6	85.0	72.7	61.5
		MCWB	46.7	48.3	52.0	54.4	59.3	62.4	65.0	64.3	62.3	56.5	50.8	46.0
	10%	DB	59.1	62.5	70.3	76.1	85.1	94.0	98.3	97.9	92.6	81.0	67.8	57.8
		MCWB	45.1	46.8	50.5	53.0	57.6	61.3	64.1	63.4	61.0	54.9	48.8	45.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.9	54.5	57.2	60.4	64.4	67.3	71.0	70.4	68.5	63.8	56.5	54.5
		MCDB	63.7	66.6	77.7	83.7	92.9	99.3	97.7	98.2	94.0	80.6	72.8	63.5
	2%	WB	51.4	52.0	55.1	57.9	62.3	65.6	69.1	68.2	66.4	60.5	54.2	51.8
		MCDB	59.6	66.1	73.9	80.2	89.7	95.8	96.1	96.4	91.5	82.1	70.4	60.3
	5%	WB	49.3	50.2	53.5	55.9	60.3	64.1	67.7	66.6	64.6	58.6	52.3	49.4
		MCDB	57.8	63.3	71.0	76.4	86.1	93.5	94.6	95.8	89.4	79.7	68.1	57.1
	10%	WB	47.2	48.5	51.7	54.1	58.8	62.4	66.2	65.0	62.7	56.7	50.6	46.8
		MCDB	56.5	59.9	67.3	73.1	82.6	90.4	93.1	94.5	87.6	76.5	65.3	55.4
Mean Daily Temperature Range	5% DB	MDBR	26.7	25.4	26.9	26.6	26.5	28.3	29.0	31.3	32.6	32.2	30.0	26.5
		MCDBR	34.6	36.3	36.6	35.6	33.5	32.8	33.1	34.7	37.1	39.2	38.2	33.1
	5% WB	MCWBR	19.9	19.0	17.2	15.5	13.3	12.2	12.3	13.7	15.3	18.2	19.9	19.5
		MCWBR	24.2	28.6	30.6	30.3	30.9	30.7	28.4	30.6	32.1	32.7	30.9	23.8
Clear-Sky Solar Irradiance	taub	0.292	0.302	0.320	0.347	0.361	0.355	0.382	0.362	0.347	0.332	0.299	0.292	
	taud	2.504	2.491	2.432	2.358	2.342	2.373	2.342	2.374	2.412	2.434	2.509	2.523	
	Ebn at Noon	293	301	303	297	292	293	284	289	289	285	287	286	
	Edh at Noon	26	30	34	39	40	39	40	38	35	31	26	24	
All-Sky Solar Radiation	RadAvg	977	1263	1723	2128	2403	2626	2504	2353	1996	1501	1118	891	
	RadStd	75	139	98	101	119	116	103	60	46	84	48	67	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	N/A	N/A	-3.46	N/A	N/A	+2.12	N/A	N/A	N/A	N/A
(47) Regional (46 neighbors)	+1.13	N/A	-3.20	+1.19	N/A	N/A	N/A	-207	+401	+191

Nomenclature: See separate page